Original Manuscript ID: Access-2023-31917

07-01-2024

Original Article Title: "Electricity Theft Detection for Smart Homes: Harnessing the Power of Machine Learning with Real and Synthetic Attacks"

To: IEEE Access Editor

Re: Response to reviewers

Dear Editor,

Thank you for allowing a resubmission of our manuscript, with an opportunity to address the reviewers' comments.

We are uploading (a) our point-by-point response to the comments (below) (response to reviewers, under "Author's Response Files*"*), (b) an updated manuscript with yellow highlighting indicating changes (as "Highlighted PDF*"*), and (c) a clean updated manuscript without highlights ("Main Manuscript"*).*

Best regards,

Abraham et al.

Reviewer#2, Concern # 1 (please list here): The contribution is very weak as thousands of relevant works are doing exactly the same using machine learning in electricity theft detection and evaluated benign and malicious samples.

Author response: We appreciate your critical assessment of our work and the opportunity to clarify the unique contributions of our study in the field of electricity theft detection using machine learning.

Clarification of Novelty

Our work is unique with the following contributions that set us apart from thousands of related works:

1. Unique Approach: "Our study introduces a novel method that leverages the aggregated power_base of appliances in smart homes. This approach is distinct in its focus on the aggregation of appliance-level data, as opposed to individual appliance data, which is the common approach in most existing studies." – Section V, subsection B (Overview of feature set)
2. Synthetic Attack Classifier: "Furthermore, our model employs a knowledge-based synthetic attack classifier to address the challenge of lack of labeled data. This is a significant deviation from traditional methods as it allows for the generation and labeling of synthetic attacks, providing a more robust mechanism for anomaly detection indicative of electricity theft." Section IV (Electricity theft detection with synthetic attack) and Section V, subsection C3
3. Contextual Relevance: "The application of our model in a smart home network environment specifically addresses the complexities and nuances of modern smart home ecosystems, which have not been the primary focus of many existing models." Literature review Table 1, highlighted our model contributions filling gaps of related works in electricity theft domain.

Highlighting the Impact

Our approach not only enhances the accuracy of theft detection, high detection rate and low false positive rare (FPR) with average AUC 98.69% and 98.74% for both simulated and real attack detection in smart home networks but also demonstrates greater adaptability to the evolving nature of smart home technologies. The use of a synthetic attack classifier, in particular, positions our model to effectively cope with new and unforeseen attack vectors, a significant advantage over more static, traditional models. Table 1.

Empirical Evidence

In our experimental evaluations, the proposed model, XGB outperformed traditional, SVM methods by 41.10% in multiclass and 5.34% in binary classification respectively in accuracy and demonstrated a higher detection rate in real-world test scenarios. These results underscore the efficacy of our approach in a rapidly advancing technological landscape. Section IV subsection E (Model performance comparison) – Table 13.

We thank you for your insights and hope that our response clarifies the unique contributions of our work. We believe that our research represents a meaningful advancement in the field of electricity theft detection within smart home networks and are committed to further refining and validating our approach.

Author action: **We have updated the manuscript by** rearranging the introduction and highlighting the sections and Tables mentioned above. We added Table 1- Literature review and gaps filled by our model in section I, subsection B.

Reviewer#2, Concern # 2 (please list here): Are the references provided applicable and sufficient? No

Author response: We added more references in literature review section 1, subsection B

Author action: We updated the manuscript by adding references in 11, 13, 14, 15, 16, 17, 18, 25, 30, 40, 46, and 47 respectively.

Reviewer#3, Concern # 1 (please list here): 1. The main motivation of the paper as well as a summary of the contribution should be stated in the abstract.

Author response: The abstract section has been reformatted; motivation is highlighted in the first paragraph while the contribution in the last paragraph.

Author action: We have updated the manuscript by reformatting the abstract. The introduction section has been updated to include the research gap, literature review, main motivation of the paper as well as a summary of the contribution in Section I

Reviewer#3, Concern # 2 (please list here): 2. It is better to divide the Introduction into the following sections.

a. The motivation for this study is as follows:

b. Literature Review:

c. Study gaps:

d. Contribution:

Author response: The motivation is at page 2 in Section I, subsection A – Challenges and motivation.

Author action: We updated the manuscript by creating Motivation, Literature review, study gaps, and contribution at pages 2 – 3 subsections A, B, and C, in the Introduction section.

Reviewer#3, Concern # 3 (please list here): 3. The structure of the paper as a whole is inconsistent with the overall review:

Author response: We have restructured the paper in the following order:

Introduction:

- Challenges and motivation
- Literature review
- Study gaps
- Contribution

The remaining section's structure is highlighted at page 3, section I, last paragraph.

Author action: We have updated the manuscript by rearranging each section in our manuscript for consistency.

Reviewer#4, Concern # 1 (please list here):

1) What is the rationale behind classifying the attacks to 5 categories shown in Fig. 1. In other words, given that the proposed framework in Fig. 3 is able to recognize the attack and also the class of the attack, what the importance of such a detection/classification will be for the operator?

Author response: The caption in Fig 3 (now Fig 2, after we updated the manuscript), page 5, was not correct and we have corrected this error. The 5-attack classification visualization in Fig 1 are the simulated real-world attack scenarios generated from appliance aggregated power consumption dataset in our previous paper [8]. This study expands the limitations of our previous work [8] to detect attacks that are not classified, hence the framework in Fig 2 (previously Fig 3) accommodates the five attacks scenarios + unknown attacks, n, as '1', as depicted in Fig 8, and the benign or non-attack data as '0' for our proposed ETD.

The importance of the detection/classification was highlighted at page 5 in section IV, paragraph 8 of our manuscript.

Author action: We updated the manuscript by correcting errors in Fig. 2 caption and also included the benefit of recognizing both specific attacks and unclassified anomalies.

Reviewer#4, Concern # 2 (please list here): 2) Please provide more insights on the model selection and feature selection for the proposed machine learning framework. I recommend including more information about maximum-likelihood parameter estimation while optimizing the objective function.

Author response: We added proposed models characteristics overview in terms of objective function optimization and maximum-likelihood parameter estimation (MLE) in page 13, Section VI, subsection A1. In selecting our model, we employed MLE to set the parameters that maximizes the likehood of observing the actual data. We also employed regularization term in the objective function of XGB, randomness in tree generation for RF, backpropagation for MLP, to mitigate the risk of overfitting, thereby enhancing our model generalization.

(i) We address the model selection in page 18, section VI, subsection D (Model Training and Test Error), subsubsection 1.

(ii) The main feature of our proposed machine learning framework is the aggregated power consumption data (**power_base**). Each entry in **power_base[day][min]** represents the aggregated power consumption for a specific category for a given day and minute. The data are structured in such a way that each day has 1440 min, and for each minute, there is aggregated power consumption information for different categories of appliances such as shown in Table 2.

Features (Independent Variables)

1. **Aggregated Power Consumption Data (power_base).**

    - **Structure:** The **power_base** dictionary contains aggregated electricity consumption data for different days and minutes.

    - **Representation:** Each entry in **power_base[day][min]** represents the aggregated power consumption for a specific category for a given day and min.

Feature Selection was addressed in section V, subsection B (overview of feature selection)

Author action: We have updated the manuscript by elaborating on the model and feature selection of our proposed ML framework with relevant references. [37], [41]

Reviewer#4, Concern # 3 (please list here): 3) Please fill the gap between confusion matrices and receiver operating characteristic curves. It is not clear how the curves represent a general view on the content of the confusion matrices. Please kindly explain.

Author response: The ROC curves at page 17, Figures 17 and 18 were updated to reflect the actual values.

The ROC curve provides a general view of the model's performance across all thresholds, giving a sense of discrimination ability. In contrast, the confusion matrix provides detailed information about the performance of our model at a specific threshold level.

In the ROC curve, the AUC for each model (XGBoost, Random Forest, MLP) provides a single measure of performance across all possible classification thresholds, summarizing the trade-off between TPR and FPR.

The confusion matrices provide a more granular view. For instance:

- The MLP for Home A simulated (Home A_sim) confusion matrix indicates that the model correctly identifies 95% of benign cases (TN), the exact value is 6655, and 89% of attack cases (TP), 6065, at a specific threshold.
- The Random Forest confusion matrix showed a high TN rate of 99%, but a lower TP rate of 84%.
- The XGBoost confusion matrix shows a similarly high TN rate of 99% and a better TP rate of 91%.

The ROC curve does not show the actual values of TP, FP, TN, and FN; rather, it shows the rate at which these values change with the different thresholds. A high AUC reflects a model that has a high TPR and low FPR across different thresholds, which generally corresponds to high values of TP and TN and low values of FP and FN in the confusion matrices at a particular operating threshold.

Author action: We updated the manuscript by updating the Figures 17 and 18. At page 16, section VI, subsection C, we updated the response above.

Reviewer#4, Concern # 4 (please list here): 4) Please report the training and testing errors of the proposed framework. How to guarantee a trade-off between these two errors?

Author response: (i) For classification tasks, errors are often reported as:

- **Training Error**: 1−accuracy on the training set
- **Test Error**: 1−accuracy on the test set

The accuracy of each set was calculated as the number of correct predictions divided by the total number of predictions. We present the report of the training and testing errors in section at page 18, section VI, subsection D. Table 12 and Figure 20 shows the comparison among our models. Figure 21(c) in page 21, shows train and error comparison of our model with other benchmark models.

(ii) To guarantee a trade-off between training and test errors, we used GridSearchCV for hyperparameter tuning and included cross-validation (5) for our proposed models such that regularization parameters L1(reg_alpha) and L2(reg_lamda) for XGBoost. For Random Forest, the parameters **min_samples_split** and **min_samples_leaf** act as a form of regularization by preventing the tree from growing too deep or complex. For MLP, **alpha** is used as an L2 penalty term, and the **max_iter** parameter is set, which is combined with the default **early_stopping= true** setting.

The detailed update is at page 19, section VI, subsection D-2.

Author action: We performed experiment on our datasets to generated training and test errors reports and updated the manuscript by adding more figures and tables.

Reviewer#4, Concern # 5 (please list here): 5) The performance of the proposed framework must be compared with other machine learning-based techniques (e.g., SVM) available in the existing literature. Please also report the limitations of the proposed framework.

Author response: In our previous paper [8], we evaluated our models using nine different classifiers, including SVM, as shown at page 21, in Figure 21 (a). However, ensemble methods, such as GB and XGB, demonstrated the best overall performance for the multi-class ETD task, with high scores in all evaluated metrics. The precision-focused RF algorithm and more nuanced MLP-4 also show promising results. In contrast, traditional algorithms, such as SVM, LR, and RRC, appear to be less effective for this particular application.

We further carried out experiments with SVM and LR binary classifiers, as shown at page 21, Figure 21 (b) our proposed XGB framework demonstrate the most balanced performance and most suitable model for ETD among the evaluated options.

We address the comparison with benchmark models at page 19, in section VI, subsection E and the limitation at page20, in section VII, subsection A.

Author action: We added more figures, Figure 21 (a-d) and Tables (13- 15) for comparison analysis and we updated the manuscript creating subsection in section VII for our model limitations at page 20 - 21.

Reviewer#4, Concern # 6 (please list here): 6) There are several typos throughout the draft (e.g., Table ??, etc.). Please do a second, more focused, proofreading before final decision can be made.

Author response: Thank you for your comments, we reviewed and revised our paper/manuscript thoroughly. We have proofread and corrected the typographical errors throughout the manuscript.

Author action: We updated the manuscript by checking with PaperPal Preflight to check for grammatical issues.

Reviewer#4, Concern # 7 (please list here): 7) Is the subject matter presented in a comprehensive manner? No. It is not.

Author response: We have updated all comments from the reviewers and have represented our manuscript according to the IEEE Access Journal guidelines.

Author action: We updated the manuscript by reorganizing our figures, tables to match relevant sections and IEEE Access Journal guidelines.

Note: *References suggested by reviewers should only be added if it is relevant to the article and makes it more complete. Excessive cases of recommending non-relevant articles should be reported to ieeeaccesseic@ieee.org*